

Shinde Aditya

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer specializing in production-grade Generative AI systems, agentic architectures, and inference optimization. Proven expertise in architecting low-latency Retrieval-Augmented Generation (RAG) pipelines, building scalable real-time multimodal interfaces, and deploying high-throughput API layers. Adept at translating complex foundation models into reliable, cost-efficient enterprise applications.

TECHNICAL SKILLS

- **AI & Machine Learning:** Large Language Models (LLMs), Agentic AI Workflows, Retrieval-Augmented Generation (RAG), Fine-Tuning (LoRA, QLoRA), Prompt Engineering, Embedding Optimization.
- **Frameworks & Libraries:** PyTorch, LangChain, LlamaIndex, Hugging Face Transformers, FastAPI, OpenCV, YOLO.
- **Databases & Vector Stores:** Pinecone, ChromaDB, Milvus, PostgreSQL, MongoDB.
- **Programming Languages:** Python, SQL, JavaScript, TypeScript
- **Cloud & Infrastructure:** AWS (EC2, S3, Bedrock), Docker, Git, GitHub Actions, Weights & Biases, MLOps Pipelines.
- **Core Competencies:** Distributed Systems, Microservices Architecture, Latency Optimization, API Orchestration, System Design.

EXPERIENCE

- **AI Engineer** Aug 2025 – Apr 2026
TechOptima.ai Hyderabad, India
 - **Architected Multimodal Streaming Pipelines:** Engineered an interactive, real-time AI Avatar system utilizing state-of-the-art Speech-to-Video (S2V) synthesis frameworks and streaming LLM outputs; achieved sub-250ms audio-visual synchronization latency for production educational platforms.
 - **Optimized Advanced RAG Implementations:** Built and deployed an enterprise-grade RAG pipeline using LlamaIndex and Pinecone; implemented hybrid search (sparse/dense), re-ranking models (Cohere), and metadata filtering, reducing hallucination rates by 35% and improving context retrieval accuracy by 42%.
 - **Designed Agentic Workflows:** Developed an asynchronous, multi-agent AI system using LangChain to automate complex data analysis tasks; integrated tool-calling loops and self-correction mechanics, increasing autonomous task completion rates to 94%.
 - **Engineered High-Throughput API Serving Layers:** Built a containerized, asynchronous FastAPI serving layer utilizing Docker and AWS EC2 to handle high-volume concurrent LLM inference requests; optimized memory allocation and request batching, improving system throughput by 55%.
 - **Automated MLOps & Evaluation Pipelines:** Established automated CI/CD deployment tracks via GitHub Actions paired with Weights & Biases for monitoring model drift and response latency, cutting production deployment time from days to minutes.
- **Machine Learning Engineer Intern** Feb 2024 – June 2024
Conquest Technology Solutions Hyderabad, India
 - **Built Automated Data & Ingestion Pipelines:** Engineered high-performance ETL data pipelines using Python and SQL to extract, clean, and vectorize massive unstructured datasets, reducing downstream embedding generation latency by 45%.
 - **Deployed Predictive Vision Models:** Configured and validated real-time object detection models using YOLO and OpenCV for spatial monitoring; optimized model edge-inference layers to maximize frame-rate stability across consumer-grade hardware.
 - **Fine-Tuned Specialized Embeddings:** Collaborated with senior engineering teams to fine-tune domain-specific sentence embeddings; executed comprehensive hyperparameter optimization routines that scaled semantic search relevance across internal databases.

PROJECTS

- **torch-secorder (DL Optimization Library) Creator / Open-Source Contributor** [GitHub](#) — [Docs](#) — [PyPI](#)
Python, PyTorch, Sequence Modeling, Poetry, CI/CD
 - **Developed Optimization Core:** Programmed a custom, PyTorch-native library executing Second-Order Optimization techniques, introducing Hessian-Vector Products (HVP) and Gauss-Newton matrix calculations to accelerate convergence bounds.
 - **Optimized Compute Profiles:** Implemented Hutchinson's trace estimation method to facilitate highly efficient curvature analysis directly on consumer-grade GPUs, cutting training overhead down by 22%.
 - **Scaled Production Adoption:** Packaged and open-sourced the library via PyPI with complete distributed training configurations and automatic parameter tuning, gaining direct traction among independent machine learning researchers.
- **Autonomous Enterprise Intelligence Platform** [GitHub](#)
Python, FastAPI, Pinecone, LangChain, Docker
 - **Engineered Multi-Agent Architecture:** Built an autonomous multi-agent system utilizing LangChain to execute complex market analysis workflows; integrated tool-use loops, dynamic vector routing, and self-correction layers.
 - **Optimized Storage & Retrieval:** Implemented a multi-tenant vector database infrastructure using Pinecone with advanced metadata filtering and hybrid lexical/semantic search, driving retrieval latency under 80ms.
 - **Streamlined Deployment:** Containerized the entire microservice ecosystem using Docker, deploying automated orchestration gates to handle high-throughput concurrent workloads efficiently.

EDUCATION

- **Bachelor of Technology in Data Science & Artificial Intelligence** Sep 2021 – Oct 2025
Woxsen University *Hyderabad, India*

ACHIEVEMENTS

- **Published Research – AI for Coral Species Classification:** Co-authored a peer-reviewed paper titled “*AI Infused Underwater Analysis for Enhanced Coral Species Classification*” IndiaCom 2025, IEEE. [🔗](#)
- **Best Presentation Award – OIST Hackathon:** Led a cross-functional team to develop a real-time language translation tool using speech-to-text APIs and transformer-based models; recognized for clarity, innovation and technical depth.
- **PyPI Author – Open-Source ML Tools:** Published multiple production-grade Python packages to PyPI (`ml-assert`, `torch-secorder`) with CI/CD pipelines, documentation, and adoption-ready CLI tools.